

Smart Bangladesh Digital City Network

A start-to-finish build plan for the six-department SBDC infrastructure in Cisco Packet Tracer — addressing, routing, services, failover proof and submission — broken into 63 tasks with story points so three people carry equal load.

BASE NETWORK	FROM STUDENT ID	TASKS	STORY POINTS	PER MEMBER	
11.8.0.0/16	22201108	63	138	46 ≈ 11.5 h	

DEVICES
38

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01 Design decisions taken before anyone opens Packet Tracer

Five judgement calls are baked into everything below. Read them first — two of them are the difference between a working failover demo and one that quietly refuses to fail over.

1. The PHD–TMA serial link is deliberately kept out of RIP

The spec asks for a direct PHD–TMA link (§2.1.3f) *and* for the failure of the SCCC–TMA path to reroute traffic through `TMA → DCS → ERU → SCCC` (§2.6.2). Those two clauses fight each other: if RIP runs over the PHD–TMA serial, then when TMA loses the central switch it simply learns everything from PHD at AD 120, the AD 130 floating static never installs, and the required failover path is never taken.

The fix is to build the cable, address both ends, keep it pingable — and mark it `passive-interface` in RIP on *both* routers. The subnet is still advertised to everyone else; no routing information crosses that link. The physical redundancy the spec asked for exists, and the failover path stays deterministic.

2. Every router-to-router link except the central segment is serial

Only SCCC–ERU is required to be serial (§2.1.3c), but making ERU–DCS, DCS–TMA, DCS–ESC and PHD–TMA serial too buys something concrete: the DCS router is required to use **directly connected static routes** (§2.5.3b) — exit-interface-only, no next hop. On an Ethernet link that form is broken by design (the router ARPs for every remote destination and leans on proxy ARP). On a point-to-point serial it is exactly correct. All six routers become 2811s with WIC-1T / WIC-2T modules.

3. SCCC redistributes its statics into RIP

§2.5.2 says SCCC runs both dynamic and static routing and must know every network. That

is only half the job — TMA and PHD are RIP-only and would never learn the ERU / DCS / ESC side. `redistribute static metric 5` under `router rip` on SCCC is what makes §2.6.1 ("TMA reaches all external networks through SCCC") true.

4. ERU reaches TMA's LAN via DCS, not via SCCC

ERU is static-only and cannot detect a failure three hops away. If ERU pointed at SCCC for `11.8.5.0/24`, then during failover SCCC would forward TMA-bound traffic to ERU and ERU would send it straight back — a two-router loop. Pointing ERU at DCS for TMA's LAN is correct in *both* states, so no loop can form. The path is mildly asymmetric during normal operation; pings and TCP are unaffected.

5. RIP timers are tuned down

When TMA's central-switch port is shut, TMA purges its RIP routes instantly (interface down) and the floating statics install immediately — that half of the demo is fast. The *return* path is not: SCCC's own port stays up (PHD is still on that switch), so SCCC must wait out the default 180 s invalid + 240 s flush timers before its floating static to TMA installs.

`timers basic 10 30 30 40` on all three RIP routers turns a four-minute wait into about forty seconds. Set it identically on SCCC, TMA and PHD or the protocol destabilises.

SERVERS STAY INSIDE DEPARTMENT LANS

§2.4.6c says to subnet server networks "if separated". Keeping DNS, web, mail and DHCP servers on their department LAN with static addresses from the top of the range satisfies §2.4.7, keeps the VLSM table readable, and removes six subnets and six router sub-interfaces of work. Every server address is documented in §04 as if it were its own allocation.

02 Topology

Six departments, six routers, one central switch, five serial WAN links. Thirty-eight devices in total: 6 routers, 7 switches, 12 PCs, 6 printers, 7 servers.

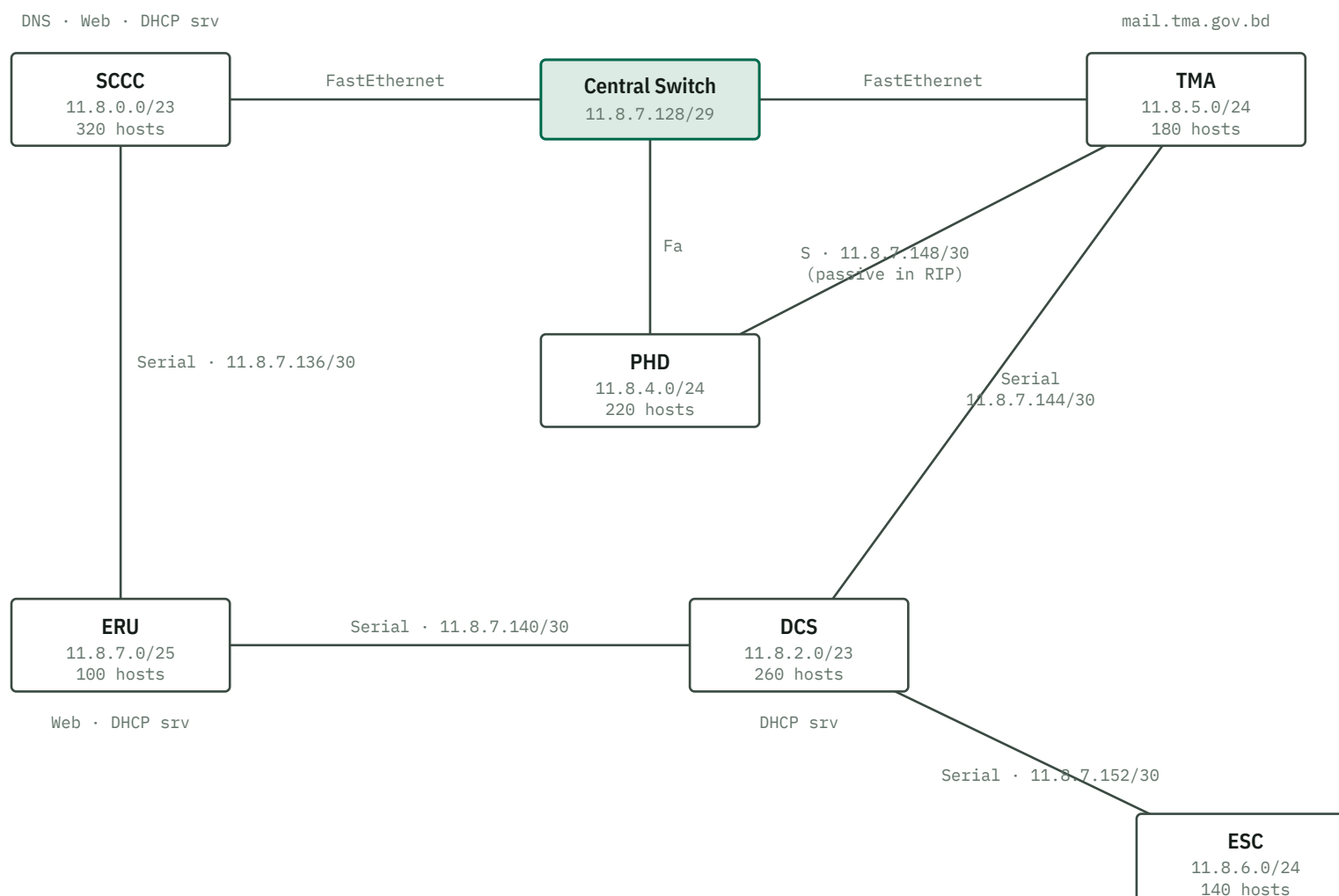


Figure 1 — WAN topology. SCCC, TMA and PHD share the central switch segment; every other link is point-to-point serial. Redundancy comes from the ring

SCCC → ERU → DCS → TMA → central switch → SCCC, which is what carries traffic when the TMA uplink drops. The PHD–TMA serial exists physically but carries no routing information, by design.

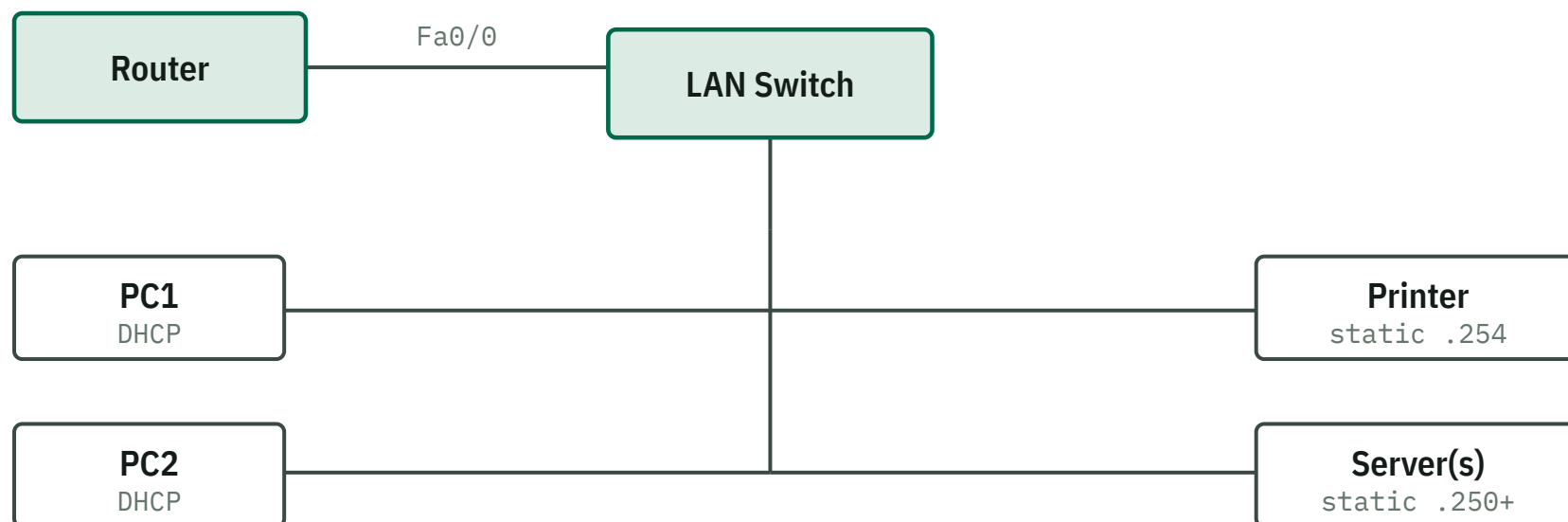


Figure 2 — The department block, repeated six times. Every department gets exactly this: one router, one LAN switch, two PCs on DHCP, one printer on a static address. SCCC additionally carries the DNS and web servers, ERU a web server and a DHCP server, DCS a DHCP server, TMA and ESC one mail server each.

03 VLSM calculation

Student ID 22201108 → last four digits 1108 → split into 11 and 08 → base network 11.8.0.0/16. The first pair is not 00, so no substitution is needed.

Allocation is largest-block-first from 11.8.0.0, so no small subnet ever blocks a large one. The six LANs consume everything up to 11.8.7.127; infrastructure links are carved out of the tail of 11.8.7.x; 11.8.8.0 onwards is untouched growth space.

STEP 1 – SIZE EACH LAN

HOST REQUIREMENT → BLOCK SIZE							
ORDER	NETWORK	HOSTS NEEDED	+2 (NET & BCAST)	BLOCK	PREFIX	USABLE	SPARE
1	SCCC LAN	320	322	512	/23	510	190
2	DCS LAN	260	262	512	/23	510	250
3	PHD LAN	220	222	256	/24	254	34
4	TMA LAN	180	182	256	/24	254	74
5	ESC LAN	140	142	256	/24	254	114
6	ERU LAN	100	102	128	/25	126	26
7	Central segment	3	5	8	/29	6	3
8–12	5 × WAN link	2	4	4	/30	2	0

WHY DCS NEEDS A /23

260 hosts is the trap in this brief. A /24 gives 254 usable addresses — six short. DCS

must take a /23, which is why it sits second in the allocation order rather than fourth by host count.

STEP 2 – ASSIGN THE BLOCKS

VLSM ALLOCATION FROM 11.8.0.0/16					
SUBNET	NETWORK	MASK	FIRST HOST	LAST HOST	BROADCAST
SCCC LAN	11.8.0.0/23	255.255.254.0	11.8.0.1	11.8.1.254	11.8.1.255
DCS LAN	11.8.2.0/23	255.255.254.0	11.8.2.1	11.8.3.254	11.8.3.255
PHD LAN	11.8.4.0/24	255.255.255.0	11.8.4.1	11.8.4.254	11.8.4.255
TMA LAN	11.8.5.0/24	255.255.255.0	11.8.5.1	11.8.5.254	11.8.5.255
ESC LAN	11.8.6.0/24	255.255.255.0	11.8.6.1	11.8.6.254	11.8.6.255
ERU LAN	11.8.7.0/25	255.255.255.128	11.8.7.1	11.8.7.126	11.8.7.127
Central segment	11.8.7.128/29	255.255.255.248	11.8.7.129	11.8.7.134	11.8.7.135
SCCC ↔ ERU	11.8.7.136/30	255.255.255.252	11.8.7.137	11.8.7.138	11.8.7.139
ERU ↔ DCS	11.8.7.140/30	255.255.255.252	11.8.7.141	11.8.7.142	11.8.7.143
DCS ↔ TMA	11.8.7.144/30	255.255.255.252	11.8.7.145	11.8.7.146	11.8.7.147
PHD ↔ TMA	11.8.7.148/30	255.255.255.252	11.8.7.149	11.8.7.150	11.8.7.151
DCS ↔ ESC	11.8.7.152/30	255.255.255.252	11.8.7.153	11.8.7.154	11.8.7.155
Reserved / growth	11.8.7.156 – 11.8.255.255	–	–	–	–

04 IP address table

Routers, printers and servers are static (§2.4.7). PCs are dynamic (§2.4.8). Every static host inside a LAN is parked at the top of its range and excluded from the DHCP pool, so the pool and the static block can never collide.

STATIC ADDRESSING – ROUTERS					
DEVICE	INTERFACE	CONNECTS TO	IP ADDRESS	MASK	NOTES
R-SCCC	Fa0/0	SCCC LAN switch	11.8.0.1	255.255.254.0	Gateway + DHCP server
	Fa0/1	Central switch	11.8.7.129	255.255.255.248	RIP active
	Se0/0/0	R-ERU Se0/0/0	11.8.7.137	255.255.255.252	DCE · clock 64000
R-TMA	Fa0/0	TMA LAN switch	11.8.5.1	255.255.255.0	Gateway + DHCP relay
	Fa0/1	Central switch	11.8.7.130	255.255.255.248	RIP active · <i>the link that fails</i>
	Se0/0/0	R-DCS Se0/0/0	11.8.7.146	255.255.255.252	Floating static next hop
	Se0/0/1	R-PHD Se0/0/0	11.8.7.149	255.255.255.252	DCE · passive in RIP
R-PHD	Fa0/0	PHD LAN switch	11.8.4.1	255.255.255.0	Gateway + DHCP relay
	Fa0/1	Central switch	11.8.7.131	255.255.255.248	RIP active
	Se0/0/0	R-TMA Se0/0/1	11.8.7.150	255.255.255.252	Passive in RIP
R-ERU	Fa0/0	ERU LAN switch	11.8.7.1	255.255.255.128	Gateway
	Se0/0/0	R-SCCC Se0/0/0	11.8.7.138	255.255.255.252	DTE
	Se0/0/1	R-DCS Se0/1/0	11.8.7.141	255.255.255.252	DCE · clock 64000

R-DCS	Fa0/0	DCS LAN switch	11.8.2.1	255.255.254.0	Gateway
	Se0/0/0	R-TMA Se0/0/0	11.8.7.145	255.255.255.252	DCE · clock 64000
	Se0/0/1	R-ESC Se0/0/0	11.8.7.153	255.255.255.252	DCE · clock 64000
	Se0/1/0	R-ERU Se0/0/1	11.8.7.142	255.255.255.252	DTE
R-ESC	Fa0/0	ESC LAN switch	11.8.6.1	255.255.255.0	Gateway + DHCP relay
	Se0/0/0	R-DCS Se0/0/1	11.8.7.154	255.255.255.252	DTE · default route exit

STATIC ADDRESSING – SERVERS AND PRINTERS						
DEPARTMENT	DEVICE	ROLE	IP ADDRESS	MASK	GATEWAY	DNS
SCCC	SRV-DNS	Central DNS	11.8.1.250	255.255.254.0	11.8.0.1	11.8.1.250
	SRV-WEB-SC	www.smartcity.gov.bd	11.8.1.251	255.255.254.0	11.8.0.1	11.8.1.250
	PRN-SCCC	Network printer	11.8.1.254	255.255.254.0	11.8.0.1	11.8.1.250
DCS	SRV-DHCP-DCS	DHCP for DCS + ESC	11.8.2.250	255.255.254.0	11.8.2.1	11.8.1.250
	PRN-DCS	Network printer	11.8.3.254	255.255.254.0	11.8.2.1	11.8.1.250
PHD	PRN-PHD	Network printer	11.8.4.254	255.255.255.0	11.8.4.1	11.8.1.250
TMA	SRV-MAIL-TMA	mail.tma.gov.bd	11.8.5.250	255.255.255.0	11.8.5.1	11.8.1.250
	PRN-TMA	Network printer	11.8.5.254	255.255.255.0	11.8.5.1	11.8.1.250
ESC	SRV-MAIL-ESC	mail.esc.gov.bd	11.8.6.250	255.255.255.0	11.8.6.1	11.8.1.250
	PRN-ESC	Network printer	11.8.6.254	255.255.255.0	11.8.6.1	11.8.1.250
	SRV-WEB-ER	www.emergency.gov.bd	11.8.7.120	255.255.255.128	11.8.7.1	11.8.1.250

ERU	SRV-DHCP-ERU	DHCP for ERU	11.8.7.121	255.255.255.128	11.8.7.1	11.8.1.250
	PRN-ERU	Network printer	11.8.7.126	255.255.255.128	11.8.7.1	11.8.1.250

DYNAMIC ADDRESSING – DHCP POOLS						
LAN	SERVED BY	POOL RANGE	GATEWAY	DNS	DELIVERY	
SCCC	R-SCCC (router pool SCCC-POOL)	11.8.0.10 – 11.8.1.199	11.8.0.1	11.8.1.250	Direct	
TMA	R-SCCC (router pool TMA-POOL)	11.8.5.10 – 11.8.5.199	11.8.5.1	11.8.1.250	Relay via ip helper-address	
PHD	R-SCCC (router pool PHD-POOL)	11.8.4.10 – 11.8.4.199	11.8.4.1	11.8.1.250	Relay via ip helper-address	
DCS	SRV-DHCP-DCS (11.8.2.250)	11.8.2.10 – 11.8.3.199	11.8.2.1	11.8.1.250	Direct	
ESC	SRV-DHCP-DCS (11.8.2.250)	11.8.6.10 – 11.8.6.199	11.8.6.1	11.8.1.250	Relay §2.3.4	
ERU	SRV-DHCP-ERU (11.8.7.121)	11.8.7.10 – 11.8.7.99	11.8.7.1	11.8.1.250	Direct	

05 Interface & hardware map

All six routers are **Cisco 2811** — two on-board FastEthernet ports plus two WIC slots. Serial ports only appear after a WIC module is fitted, and modules can only be fitted with the device powered off.

DO THIS BEFORE ANY CABLING

In Packet Tracer: click the router → **Physical** tab → click the power switch to turn it **off** → drag the WIC module into an empty slot → power back **on**. Adding a module to a running router silently fails, and the serial interfaces never appear in the CLI.

MODULES AND PORT USAGE PER ROUTER							
ROUTER	SLOT 0	SLOT 1	FA0/0	FA0/1	SE0/0/0	SE0/0/1	SE0/1/0
R-SCCC	WIC-1T	—	LAN	Central sw	→ ERU DCE	—	—
R-TMA	WIC-2T	—	LAN	Central sw	→ DCS	→ PHD DCE	—
R-PHD	WIC-1T	—	LAN	Central sw	→ TMA	—	—
R-ERU	WIC-2T	—	LAN	unused	→ SCCC	→ DCS DCE	—
R-DCS	WIC-2T	WIC-1T	LAN	unused	→ TMA DCE	→ ESC DCE	→ ERU
R-ESC	WIC-1T	—	LAN	unused	→ DCS	—	—

Exactly one end of every serial link is DCE and carries `clock rate 64000`. In Packet Tracer, pick the **Serial DCE** cable and click the DCE router *first* — the small clock icon appears on that end. Get this wrong and the line protocol stays down with no obvious error.

Device naming convention

Use these names verbatim; every command block and screenshot caption in this plan assumes them.

Routers	R-SCCC	R-TMA	R-PHD	R-ESC	R-ERU	R-DCS	
Switches	SW-SCCC	SW-TMA	SW-PHD	SW-ESC	SW-ERU	SW-DCS	SW-CENTRAL
PCs	SCCC-PC1 / SCCC-PC2 ... (same pattern per department)						
Printers	PRN-SCCC	PRN-TMA	PRN-PHD	PRN-ESC	PRN-ERU	PRN-DCS	
Servers	SRV-DNS	SRV-WEB-SC	SRV-WEB-ER	SRV-MAIL-TMA			
	SRV-MAIL-ESC	SRV-DHCP-DCS	SRV-DHCP-ERU				

06 Routing & failover design

WHO RUNS WHAT			
ROUTER	PROTOCOL	STATIC ROUTE STYLE	SPEC CLAUSE
R-SCCC	RIPv2 + static (hybrid)	Next-hop, plus one floating at AD 130, plus <code>redistribute static</code>	2.5.1, 2.5.2
R-TMA	RIPv2	Floating static via DCS at AD 130	2.5.1, 2.5.3d
R-PHD	RIPv2	None	2.5.1
R-ERU	None	Next-hop static to every remote LAN	2.5.3a
R-DCS	None	Directly connected (exit-interface only) — never fully specified	2.5.3b
R-ESC	None	One default static toward DCS	2.5.3c

The three static route forms, and which one goes where

Next-hop

```
ip route net mask 11.8.7.137
```

Names the neighbour's IP only. Used on **R-ERU** and **R-SCCC**.

Directly connected

```
ip route net mask Serial0/1/0
```

Names the exit interface only. Valid on point-to-point serial. Used on **R-DCS**.

Fully specified

```
ip route net mask Se0/1/0 11.8.7.141
```

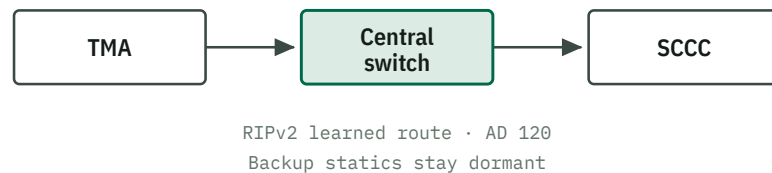
Both together. **Explicitly forbidden** on R-DCS by §2.5.3b(ii) — do not use it anywhere.

How the failover works

A floating static is just a static route with a manually raised administrative distance. RIP

has AD 120; the backup statics are configured at AD 130, so while RIP is advertising a prefix the static stays out of the routing table entirely. Shut TMA's `Fa0/1` and every RIP route TMA learned through that interface is purged at once — the AD 130 statics drop into the table in the same second.

NORMAL OPERATION



TMA UPLINK DOWN

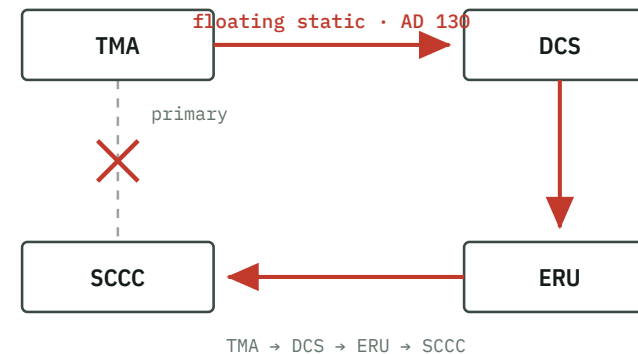


Figure 3 — What changes at the moment of failure. The only difference between the two states is which route wins for the same prefixes: a RIP entry at AD 120 through the central switch, or a static entry at AD 130 through DCS. Nothing is reconfigured during the demo — the backup is already in the running config, waiting.

Loop check

The design is loop-free in both states because two routers never point at each other for the same prefix:

- **TMA → SCCC LAN, after failure.** TMA sends to `11.8.7.145` (DCS). DCS routes `11.8.0.0/23` out `Se0/1/0` to ERU. ERU routes `11.8.0.0/23` to `11.8.7.137` (SCCC). Delivered in three hops, exactly the path §2.6.2 demands.

- **SCCC → TMA LAN, after failure.** The RIP entry is gone, so SCCC's AD 130 static sends to `11.8.7.138` (ERU). ERU's static for `11.8.5.0/24` points at DCS, not back at SCCC — this is decision 4 in §01, and it is the single thing preventing a ping-pong loop. DCS delivers over `Se0/0/0`.
- **PHD → TMA LAN, after failure.** PHD relearns `11.8.5.0/24` from SCCC, because SCCC's newly active floating static is picked up by `redistribute static`. Traffic goes PHD → SCCC → ERU → DCS → TMA. The idle PHD–TMA serial is not used, as intended.

07 Router configurations

Paste-ready blocks. Each starts from `enable` → `configure terminal`. Finish every router with `end` then `copy running-config startup-config`.

R-SCCC – HYBRID: RIPV2 + STATIC + DHCP SERVER

```
hostname R-SCCC
no ip domain-lookup
enable secret cisco123
banner motd #SBDC - Smart City Control Center - Authorized access only#
line console 0
  password cisco
  login
  logging synchronous
line vty 0 4
  password cisco
  login
exit
!--- interfaces
interface FastEthernet0/0
  description SCCC LAN
```

```
ip address 11.8.0.1 255.255.254.0
no shutdown
interface FastEthernet0/1
description Central switch segment
ip address 11.8.7.129 255.255.255.248
no shutdown
interface Serial0/0/0
description Serial to R-ERU (DCE)
ip address 11.8.7.137 255.255.255.252
clock rate 64000
no shutdown
exit
!--- DHCP server for SCCC, TMA and PHD (spec 2.3.2)
ip dhcp excluded-address 11.8.0.1 11.8.0.9
ip dhcp excluded-address 11.8.1.200 11.8.1.254
ip dhcp excluded-address 11.8.5.1 11.8.5.9
ip dhcp excluded-address 11.8.5.200 11.8.5.254
ip dhcp excluded-address 11.8.4.1 11.8.4.9
ip dhcp excluded-address 11.8.4.200 11.8.4.254
ip dhcp pool SCCC-POOL
network 11.8.0.0 255.255.254.0
default-router 11.8.0.1
dns-server 11.8.1.250
domain-name smartcity.gov.bd
ip dhcp pool TMA-POOL
network 11.8.5.0 255.255.255.0
default-router 11.8.5.1
dns-server 11.8.1.250
ip dhcp pool PHD-POOL
network 11.8.4.0 255.255.255.0
default-router 11.8.4.1
dns-server 11.8.1.250
exit
!--- dynamic routing (spec 2.5.1)
router rip
version 2
```

```
no auto-summary
network 11.0.0.0
passive-interface FastEthernet0/0
passive-interface Serial0/0/0
timers basic 10 30 30 40
redistribute static metric 5      ! feeds ERU/DCS/ESC nets to TMA + PHD
exit
!--- static routes toward the ERU / DCS / ESC side (spec 2.5.2)
ip route 11.8.7.0 255.255.255.128 11.8.7.138 ! ERU LAN
ip route 11.8.7.140 255.255.255.252 11.8.7.138 ! ERU-DCS link
ip route 11.8.2.0 255.255.254.0 11.8.7.138 ! DCS LAN
ip route 11.8.7.144 255.255.255.252 11.8.7.138 ! DCS-TMA link
ip route 11.8.7.152 255.255.255.252 11.8.7.138 ! DCS-ESC link
ip route 11.8.6.0 255.255.255.0 11.8.7.138 ! ESC LAN
!--- floating backup to TMA, used only when RIP loses the LAN
ip route 11.8.5.0 255.255.255.0 11.8.7.138 130
```

R-TMA – RIPV2 + FLOATING STATICS VIA DCS

```
hostname R-TMA
no ip domain-lookup
enable secret cisco123
banner motd #SBDC - Traffic Management Authority#
line console 0
password cisco
login
logging synchronous
line vty 0 4
password cisco
login
exit
interface FastEthernet0/0
description TMA LAN
ip address 11.8.5.1 255.255.255.0
ip helper-address 11.8.7.129      ! DHCP relay to R-SCCC (spec 2.3.2)
no shutdown
```

```
interface FastEthernet0/1
  description Central switch - PRIMARY PATH, this is the link we fail
  ip address 11.8.7.130 255.255.255.248
  no shutdown
interface Serial0/0/0
  description Serial to R-DCS - backup path
  ip address 11.8.7.146 255.255.255.252
  no shutdown
interface Serial0/0/1
  description Serial to R-PHD (DCE) - physical redundancy, no routing
  ip address 11.8.7.149 255.255.255.252
  clock rate 64000
  no shutdown
exit
router rip
  version 2
  no auto-summary
  network 11.0.0.0
  passive-interface FastEthernet0/0
  passive-interface Serial0/0/0
  passive-interface Serial0/0/1      ! keeps RIP off the PHD link - see decision 1
  timers basic 10 30 30 40
exit
!--- FLOATING STATIC ROUTES via DCS, AD 130 (spec 2.5.3d / 2.6)
ip route 11.8.0.0 255.255.254.0 11.8.7.145 130 ! SCCC LAN
ip route 11.8.4.0 255.255.255.0 11.8.7.145 130 ! PHD LAN
ip route 11.8.2.0 255.255.254.0 11.8.7.145 130 ! DCS LAN
ip route 11.8.6.0 255.255.255.0 11.8.7.145 130 ! ESC LAN
ip route 11.8.7.0 255.255.255.128 11.8.7.145 130 ! ERU LAN
ip route 11.8.7.128 255.255.255.248 11.8.7.145 130 ! central segment
ip route 11.8.7.136 255.255.255.252 11.8.7.145 130 ! SCCC-ERU link
ip route 11.8.7.140 255.255.255.252 11.8.7.145 130 ! ERU-DCS link
ip route 11.8.7.152 255.255.255.252 11.8.7.145 130 ! DCS-ESC link
```

R-PHD - RIPV2 ONLY

```
hostname R-PHD
no ip domain-lookup
enable secret cisco123
banner motd #SBDC - Public Health Department#
line console 0
  password cisco
  login
  logging synchronous
line vty 0 4
  password cisco
  login
exit
interface FastEthernet0/0
  description PHD LAN
  ip address 11.8.4.1 255.255.255.0
  ip helper-address 11.8.7.129      ! DHCP relay to R-SCCC
  no shutdown
interface FastEthernet0/1
  description Central switch segment
  ip address 11.8.7.131 255.255.255.248
  no shutdown
interface Serial0/0/0
  description Serial to R-TMA - physical redundancy, no routing
  ip address 11.8.7.150 255.255.255.252
  no shutdown
exit
router rip
  version 2
  no auto-summary
  network 11.0.0.0
  passive-interface FastEthernet0/0
  passive-interface Serial0/0/0      ! matches TMA's passive setting
  timers basic 10 30 30 40
```

R-ERU — NEXT-HOP STATIC ROUTES ONLY (SPEC 2.5.3A)

```
hostname R-ERU
no ip domain-lookup
enable secret cisco123
banner motd #SBDC - Emergency Response Unit#
line console 0
  password cisco
  login
  logging synchronous
line vty 0 4
  password cisco
  login
exit
interface FastEthernet0/0
  description ERU LAN
  ip address 11.8.7.1 255.255.255.128
  no shutdown
interface Serial0/0/0
  description Serial to R-SCCC
  ip address 11.8.7.138 255.255.255.252
  no shutdown
interface Serial0/0/1
  description Serial to R-DCS (DCE)
  ip address 11.8.7.141 255.255.255.252
  clock rate 64000
  no shutdown
exit
!--- every route names a NEXT-HOP IP, never an interface
ip route 11.8.0.0 255.255.254.0 11.8.7.137 ! SCCC LAN via SCCC
ip route 11.8.4.0 255.255.255.0 11.8.7.137 ! PHD LAN via SCCC
ip route 11.8.7.128 255.255.255.248 11.8.7.137 ! central via SCCC
ip route 11.8.7.148 255.255.255.252 11.8.7.137 ! PHD-TMA via SCCC
ip route 11.8.2.0 255.255.254.0 11.8.7.142 ! DCS LAN via DCS
ip route 11.8.6.0 255.255.255.0 11.8.7.142 ! ESC LAN via DCS
ip route 11.8.5.0 255.255.255.0 11.8.7.142 ! TMA LAN via DCS <-- loop guard
ip route 11.8.7.144 255.255.255.252 11.8.7.142 ! DCS-TMA via DCS
ip route 11.8.7.152 255.255.255.252 11.8.7.142 ! DCS-ESC via DCS
```

R-DCS – DIRECTLY CONNECTED STATIC ROUTES ONLY (SPEC 2.5.3B)

```
hostname R-DCS
no ip domain-lookup
enable secret cisco123
banner motd #SBDC - Digital Citizen Service#
line console 0
  password cisco
  login
  logging synchronous
line vty 0 4
  password cisco
  login
exit
interface FastEthernet0/0
  description DCS LAN
  ip address 11.8.2.1 255.255.254.0
  no shutdown
interface Serial0/0/0
  description Serial to R-TMA (DCE)
  ip address 11.8.7.145 255.255.255.252
  clock rate 64000
  no shutdown
interface Serial0/0/1
  description Serial to R-ESC (DCE)
  ip address 11.8.7.153 255.255.255.252
  clock rate 64000
  no shutdown
interface Serial0/1/0
  description Serial to R-ERU
  ip address 11.8.7.142 255.255.255.252
  no shutdown
exit
!--- EXIT INTERFACE only. No next-hop IP anywhere: a route carrying both
!--- would be a fully specified static, which 2.5.3b(ii) forbids.
```

```
ip route 11.8.5.0 255.255.255.0 Serial0/0/0 ! TMA LAN
ip route 11.8.7.148 255.255.255.252 Serial0/0/0 ! PHD-TMA link
ip route 11.8.6.0 255.255.255.0 Serial0/0/1 ! ESC LAN
ip route 11.8.0.0 255.255.254.0 Serial0/1/0 ! SCCC LAN
ip route 11.8.4.0 255.255.255.0 Serial0/1/0 ! PHD LAN
ip route 11.8.7.0 255.255.255.128 Serial0/1/0 ! ERU LAN
ip route 11.8.7.128 255.255.255.248 Serial0/1/0 ! central segment
ip route 11.8.7.136 255.255.255.252 Serial0/1/0 ! SCCC-ERU link
```

R-ESC - ONE DEFAULT STATIC ROUTE (SPEC 2.5.3C)

```
hostname R-ESC
no ip domain-lookup
enable secret cisco123
banner motd #SBDC - Education Service Center#
line console 0
  password cisco
  login
  logging synchronous
line vty 0 4
  password cisco
  login
exit
interface FastEthernet0/0
  description ESC LAN
  ip address 11.8.6.1 255.255.255.0
  ip helper-address 11.8.2.250 ! DHCP relay to the DCS server (spec 2.3.4)
  no shutdown
interface Serial0/0/0
  description Serial to R-DCS
  ip address 11.8.7.154 255.255.255.252
  no shutdown
exit
!--- exactly one route. Nothing else.
ip route 0.0.0.0 0.0.0.0 11.8.7.153
```

TWO MISTAKES THAT COST HOURS

Missing `no auto-summary` on any RIP router will summarise `11.8.x.x` back to the classful `11.0.0.0/8` and break every /23, /25, /29 and /30 in the design. **Mismatched** `timers basic` across SCCC, TMA and PHD makes RIP flap. All three must read `10 30 30 40`.

08 DHCP, DNS, web and mail

Server-based DHCP

Both DHCP servers are configured under **Services** → **DHCP**. Turn the service **On**, fill the pool, then press **Add** — a pool that is edited but not added is silently discarded.

SRV-DHCP-DCS AT 11.8.2.250 — TWO POOLS

POOL NAME	DEFAULT GATEWAY	DNS SERVER	START IP	SUBNET MASK	MAX USERS
DCS-POOL	11.8.2.1	11.8.1.250	11.8.2.10	255.255.254.0	300
ESC-POOL	11.8.6.1	11.8.1.250	11.8.6.10	255.255.255.0	190

SRV-DHCP-ERU AT 11.8.7.121 — ONE POOL

POOL NAME	DEFAULT GATEWAY	DNS SERVER	START IP	SUBNET MASK	MAX USERS
ERU-POOL	11.8.7.1	11.8.1.250	11.8.7.10	255.255.255.128	90

HOW THE ESC RELAY ACTUALLY RESOLVES

R-ESC receives the broadcast on Fa0/0, stamps giaddr = 11.8.6.1, and unicasts it to 11.8.2.250. The DCS server matches giaddr against its pools' gateway fields, finds ESC-P00L, and offers from 11.8.6.10 upward. Two things must be true for the reply to come back: the DCS server's own **Default Gateway must be set to 11.8.2.1**, and R-DCS must hold the static route for 11.8.6.0/24 out Serial0/0/1. Both are already in this plan — but a missing default gateway on the server device is the usual reason this silently fails.

DNS — SRV-DNS at 11.8.1.250

Services → DNS → On, then add each record and press Add.

ZONE RECORDS			
NAME	TYPE	ADDRESS	WHY IT EXISTS
www.smartcity.gov.bd	A	11.8.1.251	Required by §2.2.2a
www.emergency.gov.bd	A	11.8.7.120	Required by §2.2.2b
mail.tma.gov.bd	A	11.8.5.250	Mail client server field
mail.esc.gov.bd	A	11.8.6.250	Mail client server field
tma.gov.bd	A	11.8.5.250	Cross-domain SMTP relay
esc.gov.bd	A	11.8.6.250	Cross-domain SMTP relay
smartcity.gov.bd	A	11.8.1.251	Convenience
emergency.gov.bd	A	11.8.7.120	Convenience

THE BARE-DOMAIN RECORDS ARE NOT OPTIONAL

Packet Tracer has no MX record type. When the TMA mail server is handed a message for `student@esc.gov.bd`, it strips the domain and does a plain **A lookup on** `esc.gov.bd` to find where to relay. Without records 5 and 6 above, mail inside each domain works perfectly and cross-domain mail fails with a vague send error — the single most common way §2.2.3e is lost.

For the same reason, both **mail servers themselves must have DNS server** `11.8.1.250` set in their own IP configuration. A server that cannot resolve cannot relay.

Web servers

Services → HTTP → On (leave HTTPS on too), then edit `index.html`. Give each site real content so the screenshots read clearly:

```
<!-- SRV-WEB-SC : www.smartcity.gov.bd -->
<html><body>
<h1>Smart City Control Center</h1>
<p>Smart Bangladesh Digital City – central services portal.</p>
<p>Server 11.8.1.251 · SCCC LAN 11.8.0.0/23</p>
</body></html>

<!-- SRV-WEB-ER : www.emergency.gov.bd -->
<html><body>
<h1>Emergency Response Unit</h1>
<p>24/7 emergency coordination – Smart Bangladesh Digital City.</p>
<p>Server 11.8.7.120 · ERU LAN 11.8.7.0/25</p>
</body></html>
```

Mail servers and accounts

On each mail server: **Services → EMAIL** switch **SMTP** and **POP3** to On, set the **Domain**

On each mail server, **SERVICES** → **EMAIL**, switch **SMTP** and **POP3** to **On**, set the **Domain**

Name, then add users and press **+** for each.

MAIL ACCOUNTS — PASSWORD CISCO123 THROUGHOUT				
SERVER	DOMAIN NAME	USER	FULL ADDRESS	CLIENT PC
SRV-MAIL-TMA 11.8.5.250	tma.gov.bd	traffic1	traffic1@tma.gov.bd	TMA-PC1
		admin	admin@tma.gov.bd	TMA-PC2
SRV-MAIL-ESC 11.8.6.250	esc.gov.bd	student	student@esc.gov.bd	ESC-PC1
		teacher	teacher@esc.gov.bd	ESC-PC2

On each client PC, open **Desktop** → **Email** and fill the configuration wizard. Use host names, not IP addresses — that is what proves DNS is doing its job:

MAIL CLIENT CONFIGURATION — TMA-PC1 SHOWN, OTHERS FOLLOW THE SAME PATTERN		
FIELD	TMA-PC1	ESC-PC1
Your name	Traffic Officer 1	ESC Student
Email address	traffic1@tma.gov.bd	student@esc.gov.bd
Incoming server	mail.tma.gov.bd	mail.esc.gov.bd
Outgoing server	mail.tma.gov.bd	mail.esc.gov.bd
User / password	traffic1 / cisco123	student / cisco123

The acceptance test for §2.2.3e: send `traffic1@tma.gov.bd → student@esc.gov.bd`, press **Receive** on ESC-PC1, then reply and press **Receive** on TMA-PC1. Both directions must be captured.

09 Failover demonstration

Run this as a scripted sequence and screenshot every numbered step. It is the deliverable most likely to be examined closely, and it is worth rehearsing once before capturing.

STEP 1 – BASELINE, BEFORE TOUCHING ANYTHING

```
! on R-TMA
show ip route
! expect: R 11.8.0.0/23 [120/1] via 11.8.7.129, FastEthernet0/1
!          R 11.8.2.0/23 [120/6] via 11.8.7.129 (redistributed, metric 5+1)
! there must be NO 'S' route to those prefixes yet

show ip route 11.8.0.0
show ip protocols

! on TMA-PC1 command prompt
tracert 11.8.1.251
! expect 2 hops: 11.8.5.1 -> 11.8.7.129 -> 11.8.1.251
```

STEP 2 – BREAK THE PRIMARY PATH

```
! on R-TMA
configure terminal
interface FastEthernet0/1
shutdown
end
```

Deleting the cable between R-TMA and SW-CENTRAL in the workspace works equally well and looks more convincing in a screenshot. Either is acceptable; be consistent between the

demo and the report.

STEP 3 – PROVE THE FLOATING STATIC INSTALLED

```
! on R-TMA, immediately
show ip route
! expect: S 11.8.0.0/23 [130/0] via 11.8.7.145
!          S 11.8.2.0/23 [130/0] via 11.8.7.145
! the AD in brackets flipping 120 -> 130 IS the evidence. Screenshot this.

show ip route static
```

WAIT ABOUT 40 SECONDS BEFORE TESTING END-TO-END

TMA switches over instantly, but SCCC's return route to `11.8.5.0/24` is a RIP entry that has to age out first. With `timers basic 10 30 30 40` that takes roughly 30–40 s. Pings during that window will fail — that is expected, not a bug. If you skipped the timer tuning, use Packet Tracer's **Fast Forward Time** button (each press advances 30 s) until `show ip route` on R-SCCC shows `S 11.8.5.0/24 [130/0] via 11.8.7.138`.

STEP 4 – PROVE THE TRAFFIC REALLY TAKES TMA → DCS → ERU → SCCC

```
! on TMA-PC1 command prompt
tracert 11.8.1.251

! expected hops, and this is the money shot for spec 2.6.2:
!  1  11.8.5.1      R-TMA   (LAN gateway)
!  2  11.8.7.145    R-DCS   (arrived on Se0/0/0)
!  3  11.8.7.141    R-ERU   (arrived on Se0/0/1)
!  4  11.8.7.137    R-SCCC  (arrived on Se0/0/0)
!  5  11.8.1.251    destination

ping 11.8.1.251
```

```
ping www.smartcity.gov.bd
! and confirm the website still loads in Desktop -> Web Browser
```

STEP 5 – RESTORE AND CONFIRM RECONVERGENCE

```
! on R-TMA
configure terminal
interface FastEthernet0/1
  no shutdown
end
! wait ~20 s, then
show ip route
! the 'S ... [130/0]' entries must be replaced by 'R ... [120/x]' again
! the floating statics are still in running-config, just dormant
```

Include the before/after `show ip route` pair side by side in the report. The single most persuasive piece of evidence is the same prefix appearing first as `[120/1]` and then as `[130/0]`.

10 Task board & story points

Sixty-three tasks across nine phases, 138 points total. The phases are a genuine dependency order — you cannot verify DHCP before interfaces are addressed, and you cannot demonstrate failover before routing converges — so work down the table rather than cherry-picking.

1 POINT

About 15 minutes. A single command block, one device, nothing to work out

2 POINTS

About 30–40 minutes. Several devices, or one device plus verification

3 POINTS

About 60–90 minutes. Needs design thinking, cross-checking, or touches every

Fill the **Member** column from the suggested split in §11, or reassign freely — just keep each person's total at 46.

MASTER TASK LIST — 63 TASKS, 138 POINTS					
ID	TASK	DONE WHEN	NEEDS	PTS	MEMBER
PHASE 0 · KICKOFF & GROUND RULES					4 PTS
T01	Lock the base network from the first member's Student ID (22201108 → 11.8.0.0/16) and record the derivation rule	Base network and the two-digit split written into the report's addressing section	—	1	_____
T02	Agree PT version, file naming (SBDC_v1.pkt ...), shared drive folder, and a one-editor-at-a-time rule	Folder created and naming convention posted to the group chat	—	1	_____
T03	Build the requirement traceability checklist — one row per numbered clause in §2 of the brief	Checklist file with 40+ rows, every row marked Not started	—	2	_____
PHASE 1 · DESIGN ON PAPER					16 PTS
T04	VLSM: size and allocate the six department LANs, largest block first	Table giving block, mask, first/last host and broadcast for all six LANs	T01	3	_____
T05	VLSM: allocate the central-switch /29 and the five WAN /30s from the tail of the space	Six infrastructure subnets listed, checked for zero overlap with the LANs	T04	2	_____
T06	Build the master IP address table — every router interface, printer, server and DHCP range	One table covering all 16 router interfaces, 6 printers, 7 servers, 6 pools	T05	3	_____
T07	Interface and hardware map: which port on which router, which WIC modules, which end is	Port map table plus a module shopping list per router	T06	2	_____

DCE					
T08	Draw the labelled logical topology — departments, links, link types, subnets	Diagram exported as PNG, readable at report size	T06	3	-----
T09	Write the routing design note: who runs RIP, who is static, the AD values, and a hop-by-hop failover trace	One page showing the failover path is loop-free in both directions	T07	3	-----
PHASE 2 · BUILD THE TOPOLOGY IN PACKET TRACER				15 PTS	
T10	Place and rename 6 routers; power each off, fit WIC-1T / WIC-2T per the map, power back on	<code>show ip interface brief</code> lists the expected serial ports on all six	T07	3	-----
T11	Place and rename the 6 LAN switches plus SW-CENTRAL	Seven switches on the canvas with the agreed names	T10	2	-----
T12	Place 12 PCs, 6 printers and 7 servers; rename every single device	38 devices on the canvas, no default names left anywhere	T11	3	-----
T13	Cable all six LAN segments: router Fa0/0 → switch, switch → PCs, printer and servers	Every LAN link shows green on both ends	T12	2	-----
T14	Cable the central segment: SCCC, TMA and PHD Fa0/1 into SW-CENTRAL	Three links up on SW-CENTRAL	T13	1	-----
T15	Cable the five serial WAN links, clicking the DCE router first on each	Clock icon on the correct end of all five; DCE side matches the port map	T14	2	-----
T16	Cosmetics: text label on every device and link, colour the backup path, save the baseline <code>.pkt</code>	Topology is self-explanatory in a screenshot with no verbal explanation	T15	2	-----
PHASE 3 · ADDRESSING				17 PTS	
T17	R-SCCC: hostname, banner, console/enable/vty passwords, three interfaces, clock rate on Se0/0/0	All three interfaces up/up in <code>show ip int brief</code>	T15	2	-----

T18	R-TMA: hostname, security baseline, four interfaces, clock rate on Se0/0/1	All four interfaces up/up	T15	2	-----
T19	R-PHD: hostname, security baseline, three interfaces	All three interfaces up/up	T15	2	-----
T20	R-ESC: hostname, security baseline, two interfaces	Both interfaces up/up	T15	1	-----
T21	R-ERU: hostname, security baseline, three interfaces, clock rate on Se0/0/1	All three interfaces up/up	T15	2	-----
T22	R-DCS: hostname, security baseline, four interfaces, clock rate on Se0/0/0 and Se0/0/1	All four interfaces up/up	T15	2	-----
T23	Static IP, mask, gateway and DNS on all six network printers	Each printer pings its own gateway	T17–T22	2	-----
T24	Static IP, mask, gateway and DNS on all seven servers	Each server pings its own gateway; DNS field set to 11.8.1.250 on every one	T17–T22	2	-----
T25	Link-by-link verification: ping across every /30 and across the /29; fix any interface still down	All 6 point-to-point links and the central segment verified in both directions	T23, T24	2	-----
PHASE 4 • DHCP				13 PTS	
T26	R-SCCC: six exclusion ranges plus the three router DHCP pools (SCCC, TMA, PHD)	<code>show ip dhcp pool</code> lists three pools with correct networks	T25	3	-----
T27	<code>ip helper-address 11.8.7.129</code> on R-TMA Fa0/0 and R-PHD Fa0/0	TMA and PHD PCs receive addresses from the SCCC router pools	T26	2	-----
T28	SRV-DHCP-DCS: turn the service on, build DCS-POOL and ESC-POOL, press Add on both	Both pools listed in the service table with correct gateways and masks	T25	3	-----
T29	<code>ip helper-address 11.8.2.250</code> on R-ESC	ESC PCs receive 11.8.6.x addresses	T28	1	-----

Fa0/0		sourced from the DCS server			
T30	SRV-DHCP-ERU: turn the service on, build ERU-POOL with the /25 mask	Pool listed with mask 255.255.255.128	T25	2	-----
T31	Switch all 12 PCs to DHCP and verify with <code>ipconfig /all</code> in every department	12 PCs hold an address, gateway and DNS 11.8.1.250 from the right pool	T27, T29, T30	2	-----
PHASE 5 · ROUTING					23 PTS
T32	RIPv2 on SCCC, TMA and PHD: version 2, <code>no auto-summary</code> , <code>network 11.0.0.0</code> , passive interfaces, tuned timers	<code>show ip protocols</code> matches on all three; PHD and TMA see each other's LANs	T25	3	-----
T33	R-SCCC: the six next-hop static routes covering the ERU, DCS and ESC side	SCCC can ping every remote LAN gateway	T32	3	-----
T34	R-SCCC: <code>redistribute static metric 5</code> under <code>router rip</code>	TMA and PHD show <code>R</code> routes to 11.8.2.0/23, 11.8.6.0/24 and 11.8.7.0/25	T33	2	-----
T35	R-SCCC: floating static to the TMA LAN via ERU at AD 130	Route present in running-config, absent from <code>show ip route</code> while RIP is healthy	T34	2	-----
T36	R-ERU: the nine next-hop static routes — four toward SCCC, five toward DCS	Every route uses a next-hop IP; TMA's LAN points at DCS, not SCCC	T25	3	-----
T37	R-DCS: the eight exit-interface static routes — no next-hop IP, no fully specified routes	<code>show run include ip route</code> shows interface names only	T25	3	-----
T38	R-ESC: the single default static route toward DCS	Exactly one <code>ip route</code> line in the config	T25	1	-----
T39	R-TMA: the nine floating static routes via DCS at AD 130	All nine in running-config; none in the routing table while Fa0/1 is up	T32	3	-----
T40	Full reachability sweep: each department's PC1 to every other LAN gateway, printer and server	Completed 6×6 ping matrix with zero failures	T31, T36–T39	3	-----

PHASE 6 • SERVERS & SERVICES						17 PTS
T41	SRV-DNS: enable DNS and add all eight A records, including the bare <code>tma.gov.bd</code> and <code>esc.gov.bd</code>	Every PC resolves all four service names with <code>nslookup</code>	T40	3	-----	
T42	SRV-WEB-SC: enable HTTP and author the <code>www.smartcity.gov.bd</code> index page	Page loads by IP from a SCCC PC	T40	2	-----	
T43	SRV-WEB-ER: enable HTTP and author the <code>www.emergency.gov.bd</code> index page	Page loads by IP from an ERU PC	T40	2	-----	
T44	SRV-MAIL-TMA: SMTP and POP3 on, domain <code>tma.gov.bd</code> , users traffic1 and admin	Both users listed on the server	T41	2	-----	
T45	SRV-MAIL-ESC: SMTP and POP3 on, domain <code>esc.gov.bd</code> , users student and teacher	Both users listed on the server	T41	2	-----	
T46	Configure the four PC mail clients using host names rather than IP addresses	All four clients save without error and can send within their own domain	T44, T45	2	-----	
T47	Cross-domain email test in both directions between TMA and ESC	Message received at ESC, reply received back at TMA, both captured	T46	2	-----	
T48	Browse both websites by domain name from a PC in all six departments	12 successful page loads, 2 per department	T41–T43	2	-----	
PHASE 7 • BACKUP ROUTING DEMONSTRATION						9 PTS
T49	Capture the before state: <code>show ip route</code> on TMA and SCCC, plus <code>tracert</code> from TMA-PC1	Screenshots showing <code>[120/x]</code> entries and a 2-hop trace through the central switch	T48	2	-----	
T50	Trigger the failure by shutting R-TMA Fa0/1, and record the moment	Interface down confirmed on both the CLI and the topology view	T49	2	-----	

T51	Prove the takeover: <code>show ip route</code> showing AD 130 statics, then <code>tracert</code> showing TMA → DCS → ERU → SCCC	4-hop trace captured plus a successful ping to www.smartcity.gov.bd	T50	3	-----
T52	Restore the link and confirm RIP reconverges and the static goes dormant again	<code>[130/0]</code> entries replaced by <code>[120/x]</code> ; end-to-end ping still passes	T51	2	-----
PHASE 8 · DOCUMENTATION & SUBMISSION				24 PTS	
T53	Write the VLSM calculation section — the method and the working, not just the results	Reader can reproduce every block from the Student ID alone	T05	3	-----
T54	Write the IP address table section for the report	Routers, printers, servers and DHCP ranges all present and matching the <code>.pkt</code>	T06	2	-----
T55	Export <code>show running-config</code> from all six routers into the report appendix	Six configs, trimmed of noise, labelled by hostname	T52	2	-----
T56	Screenshot set A — DHCP address assignment: router pool, server pool and relay-sourced lease	At least four images covering all three delivery mechanisms	T31	2	-----
T57	Screenshot set B — DNS resolution and both websites loading by domain name	<code>nslookup</code> output plus both rendered pages	T48	2	-----
T58	Screenshot set C — cross-domain email, both directions	Sent and received views on both TMA and ESC clients	T47	2	-----
T59	Screenshot set D — backup route after primary link failure	Before/after routing tables side by side plus the 4-hop trace	T52	2	-----
T60	Screenshot set E — ping tests between all departments	Completed ping matrix, every cell evidenced	T40	2	-----
T61	Assemble the final report: cover page, sections matching the ten deliverables, appendix	Single PDF, every deliverable in the brief has its own numbered section	T53–T60	3	-----

T62	QA pass against the T03 traceability checklist; fix every gap found	Every clause of §2 marked Done with a pointer to where it is evidenced	T61	3	-----
T63	Package the .pkt and report, save one final time, submit	Both files uploaded before the deadline	T62	1	-----

11 The three-way split

Three tracks of exactly **46 points** each. The split is not arbitrary: each track owns two routers end to end, so one person configures a device, verifies it, and documents it — nobody debugs someone else's half-finished work. Each track also carries one design task, one verification task and one documentation task, so no one gets stuck with only screenshots.

TRACK A · 46 POINTS · 19 TASKS

Addressing & dynamic routing

Owens **R-SCCC** and **R-PHD**. The hardest single job in the project: the VLSM plan everyone else builds on, RIPv2, the redistribution that makes §2.6.1 true, and the final QA pass.

T01 T04 T05 T06 T09 T17 T19 T25 T26 T27 T

MEMBER -----

TRACK B · 46 POINTS · 23 TASKS

Topology build & static routing

Owens **R-DCS** and **R-ESC**. Builds and cables the entire workspace, then configures the three trickiest static-route forms and runs the failover demonstration.

T02 T07 T10 T11 T12 T13 T14 T15 T16 T20 T

MEMBER -----

TRACK C · 46 POINTS · 21 TASKS

Services & documentation

Owens **R-TMA** and **R-ERU**. Every server in the network — DNS, both websites, both mail servers, the ERU DHCP server — plus the topology diagram and the report assembly.

T03 T08 T18 T21 T24 T30 T36 T41 T42 T43 T

MEMBER -----

WHERE THE TRACKS HAVE TO SYNC

After Phase 1, Track A's address table has to be signed off by B and C before anyone

After Phase 1 — Track A's address table has to be signed off by B and C before anyone types an IP, because every later task depends on it.

After T15 — Track B hands over a cabled, saved `.pkt`. From here on only one person edits the file at a time; the others work from a copy and paste their configs in during their slot.

Before Phase 7 — T40 (Track A) must pass before Track B starts the failover demo. Demonstrating failover on a network that was never fully reachable proves nothing.

If your group works in one sitting instead

Ignore the tracks and work down the phases together, with one person driving Packet Tracer and the other two reading the config blocks and checking output. Slower in wall-clock time, but Phases 3 to 5 catch typos far faster with three sets of eyes. The points still record who did what.

12 Submission checklist

The ten deliverables from §3 of the brief, each mapped to the task that produces it.

DELIVERABLE → PRODUCING TASK			
#	DELIVERABLE	PRODUCED BY	WHERE IT LIVES
1	Cisco Packet Tracer file	T63	<code>SBDC_final.pkt</code>
2	Properly labelled topology	T08, T16	Report §2 + the <code>.pkt</code> itself
3	VLSM calculation	T04, T05, T53	Report §3
4	IP address table	T06, T54	Report §4

5	DHCP configurations	T26–T30	Report §5 + appendix
6	DNS configuration	T41	Report §6
7	Email server configuration	T44–T46	Report §7
8	Static routing configuration	T33, T35–T39	Report §8 + appendix
9	Dynamic routing configuration	T32, T34	Report §9 + appendix
10a	Screenshot — cross-domain email	T58	Report §10
10b	Screenshot — website access by DNS	T57	Report §10
10c	Screenshot — DHCP address assignment	T56	Report §10
10d	Screenshot — backup route after link failure	T59	Report §10
10e	Screenshot — ping tests between all departments	T60	Report §10

Final gate — walk this list before submitting

- ☐ All 12 PCs hold DHCP addresses from the correct pool, and ESC's came through the relay from DCS.
- ☐ Every PC resolves `www.smartcity.gov.bd` and `www.emergency.gov.bd` and loads both pages.
- ☐ `traffic1@tma.gov.bd` and `student@esc.gov.bd` have exchanged mail in both directions.
- ☐ R-DCS's config contains no next-hop IP on any `ip route` line — grep it and confirm.
- ☐ R-ESC's config contains exactly one `ip route` line.
- ☐ R-ERU's routes all use next-hop IPs, and `11.8.5.0/24` points at DCS, not SCCC.

- ☐ R-SCCC's routing table contains both `R` and `S` entries, and covers every network in §03.
- ☐ The failover trace shows four hops through DCS and ERU, not two through the central switch.
- ☐ The link was restored and RIP reconverged after the demo — submit the network in its working state.
- ☐ Every device on the canvas has its agreed name; no `Router0` or `PC5` survives.
- ☐ Final save done *after* the last configuration change. Reopen the file once to confirm it saved.